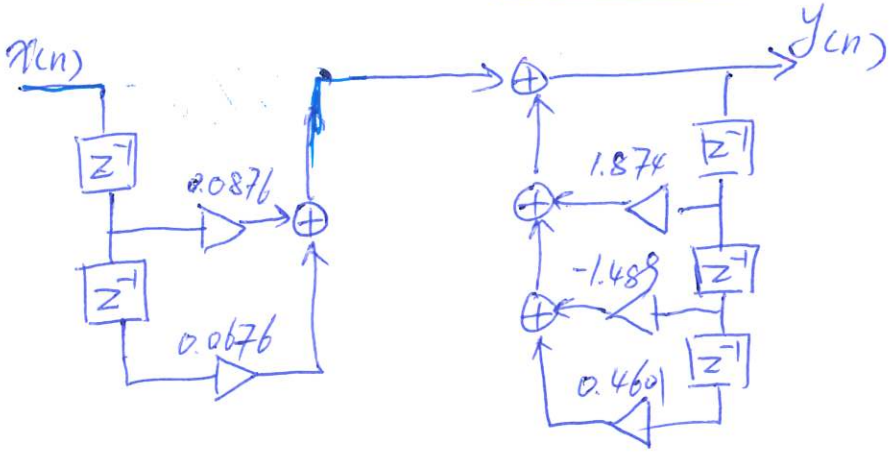
Exercise 7 Realization

1. Draw type I block diagram.

$$H(z) = \frac{Y(z)}{X(z)} \Rightarrow \boxed{\begin{aligned} &y(n) - 1.874\,y(n-1) + 1.489\,y(n-2) - 0.4601\,y(n-3) \\ &= 0.0876\,x(n-1) + 0.0676\,x(n-2) \end{aligned}}$$



we need 5 delays

2. Determine poles & zeros

$$H(z) = \frac{0.0876z^2 + 0.0676z}{z^3 - 1.874z^2 + 1.489z - 0.4601} = \frac{0.0876z\left(z + \frac{0.0676}{0.0876}\right)}{(z-0.679)(z-0.5975-0.5662i)\cdot(z-0.5975+0.5662i)}$$

zeros $z_1 = -0.7717$

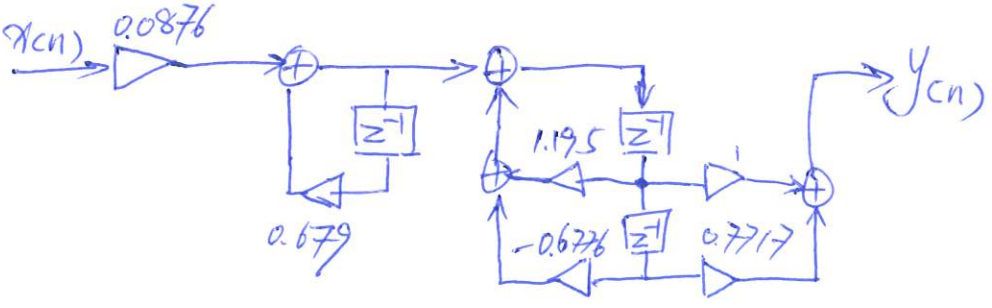
poles $p_1 = 0.679$, $p_2 = 0.5975 + 0.5662i$, $p_3 = 0.5975 - 0.5662i$

$|p_2| = |p_3| = 0.8232 < 1$ (stable!)

3. Cascade Realization

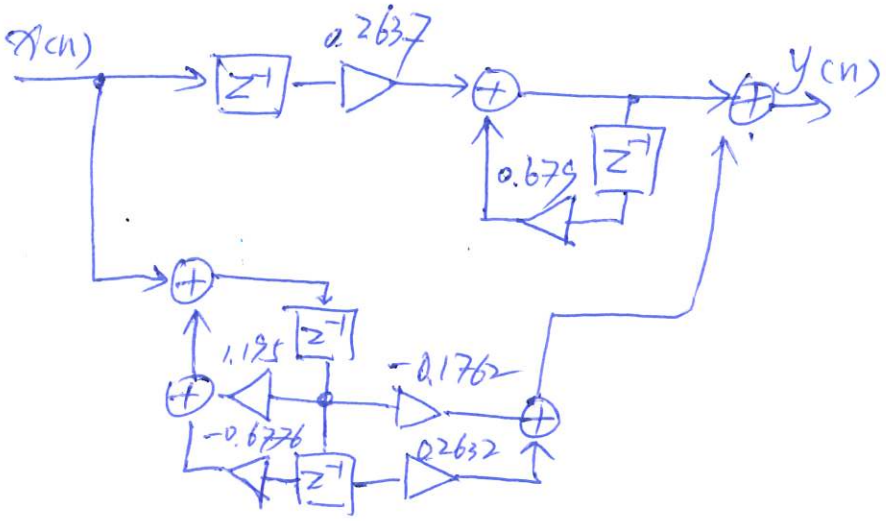
$$H(z) = 0.0876 \cdot \frac{z}{z-0.679} \cdot \frac{z+0.7717}{(z-0.5975-0.5662i)\cdot(z-0.5975+0.5662i)}$$

$$= 0.0876 \cdot \frac{1}{1-0.679z^{-1}} \cdot \frac{z^{-1} + 0.7717z^{-2}}{1 - 1.195z^{-1} + 0.6776z^{-2}}$$



4. parallel. Realization

$$H(z) = \frac{0.2637}{z - 0.679} + \frac{-0.1762z + 0.2632}{z^2 - 1.195z + 0.6776}$$



5. Low pass filter